

N=29 Murty-Simon candidate: verification companion

Reviewer-v4 dependency map, regressions, and audit boundaries

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11 September 2026. Research directed by Paul Lenz; mathematical development, implementation and internal checking by ChatGPT/Geeps.

Status: candidate mathematics. Independent mathematical review, independent computational reproduction and novelty assessment remain OPEN.

1. What changed in v4

Reviewer-v3 made the corrected minimal trusted kernel proof-critical for $\Delta=16$ at 210 and 211 edges and used progressively smaller finite checks higher in the same degree branch.

Reviewer-v4 replaces **all** proof-critical $\Delta=16$ computation at every $m \geq 210$ by the hand threshold-tail inequality

$$Q(s) \leq 18,$$

which combines with the bridge lower bound

$$Q(s) \geq 16 + 2t, \quad t=m-208,$$

to force $t \leq 1$. Therefore $\Delta=16$ is impossible whenever $m \geq 210$.

The theorem-level candidate statement is unchanged:

$$e(G) \leq 210,$$

with equality exactly $K(14,15)$.

2. Current proof architecture

Scope	reviewer-v4 method	Proof-critical computation?
$\Delta \leq 14$	degree sum	no
$\Delta = 15$	witness-deficit hand proof; equality forces $K(14,15)$	no
$\Delta = 16$, every $m \geq 210$	bridge + threshold capacity + hand tail-deficit bound	no
$\Delta = 17$	pointwise charging hand bound	no
$\Delta = 18, \dots, 27$	residual h-index hand bound	no

Scope	reviewer-v4 method	Proof-critical computation?
Delta = 28	universal-vertex/star observation	no

Thus the current N29 candidate proof is logically computation-free. Exact programs remain independent regression evidence.

3. Delta=16 trust boundary

The frozen self-contained bridge is

`project/reviews/n29/2026-09-11-reviewer-v3/GRAPH_TO_MODEL_BRIDGE.md`.

Reviewer-v4 uses only these Delta=16 consequences:

1. $S \geq r + 2t$;
 2. residual activity $\rho_u \geq 1$ for $t > 0$;
 3. selected-incidence forcing $s_i \leq \rho_u$;
 4. a selected source has $\rho_u \leq 11$, so positive $s_i \leq 11$;
 5. threshold capacity
- $$2W_h \leq z_h^2 - z_h + h(h+1).$$

It does not use the isolated-C lemma, grouped LP embedding or Farkas checker semantics.

4. Threshold-tail reduction

For $h \geq 2$, define

$$\begin{aligned} W_h &= \sum_{\{i: s_i \geq h\}} s_i, \\ g_h(W) &= \text{least threshold-compatible } z_h, \\ Q(s) &= S - \sum_{h=2}^{11} g_h(W_h). \end{aligned}$$

Residual activity turns $r = \sum \rho_u$ into a tail sum and yields

$$Q(s) \geq 16 + 2t.$$

The hand proof then writes

$$\begin{aligned} N_h &= \#\{i: s_i \geq h\}, \\ p &= N_1, \\ D &= \sum_{h=2}^{11} (N_h - g_h(W_h)), \\ Q &= p + D. \end{aligned}$$

It proves $D \leq 6$ by monotone clipping:

$$\text{all demands } > 6 \rightarrow 6 \rightarrow 5 \rightarrow 4 \rightarrow 3,$$

with explicit endpoint cases, then reduces to

$x=N_2,$
 $y=N_3,$
 $D=x+y-g_2(2x+y)-g_3(3y),$
 $0 \leq y \leq x \leq 12,$

and proves $D \leq 6$ in three elementary ranges of y . Since $p \leq 12$, $Q \leq 18$.

Detailed proof:

project/research/general_n/2026-09-11-threshold-tail-collapse-v1/N29_DELTA16_THRESHOLD_TAIL_HA

5. Independent exact regressions

The hand argument is not dependent on these computations, but two exact checks independently support it.

Local hand-obligation checker

`n29_delta16_threshold_tail_hand_check.py` checks only the finite endpoint tables used by the clipping proof and the final 91 pairs $0 \leq y \leq x \leq 12$. It uses integer arithmetic only, no solver and no floating point.

Full demand-domain checker

`n29_delta16_threshold_tail_collapse_exact.py` exhausts all

$C(23, 12) = 1,352,078$

nondecreasing demand multisets $0 \leq s_i \leq 11$ and independently finds

$\max Q(s) = 18$.

The four maximisers are

$(2, 3^{11}), (3^{12}), (3, 4^{11}), (4^{12})$.

Both checks passed together in GitHub Actions run 34617844004.

A separate rowwise audit regenerated the historical $t=3$ minimal kernel and found that all 126 residual rows fail the written threshold family, including all 94 positive zero-slack RX-Hall profiles. Run 34616219528 passed. This is corroboration only.

6. Historical minimal kernel: retained, not proof-critical

The corrected reviewer-v3 route remains valuable independent evidence:

Edge count	t	Retained demands	Residual rows	Exact Farkas rejections	Final survivors
211	3	72	126	126	0
210	2	367	1,467	1,467	0

The v1 grouped cumulative-threshold builder had a real label-group multiplicity bug and its certificates remain quarantined. Corrected v2 and exact integer Farkas verification remain preserved. Reviewer-v4 simply no longer needs that stack logically.

7. Why this reduces correlated-error risk

Reviewer-v3's largest correlated risk was the actual-graph-to-averaged-variable embedding of the corrected late LP. Reviewer-v4 removes that model from the theorem proof completely.

The highest-value independent targets are now:

1. complement/quasi-edge and selected/residual semantics;
2. residual activity;
3. selected-incidence demand forcing;
4. threshold capacity;
5. the hand tail-deficit clipping proof;
6. Delta=15 witness/equality counting;
7. Delta=17 pointwise charging and Delta>=18 residual h-index;
8. the dominating-edge citation.

8. Recommended review order

A hostile reviewer should:

1. reconstruct the selected/residual bridge from the graph;
2. verify the exact ledger $S > r + 2t$;
3. attack residual activity for collisions or hidden selected edges;
4. verify $s_i \leq \rho_u$ and $\rho_u \leq 11$ at selected sources;
5. independently prove threshold capacity;
6. independently check each clipping step in $D \leq 6$;
7. verify the final two-level inequalities;
8. audit the short Delta=15,17 and h-index branches;
9. only then use the exact regressions as corroboration.

No rerun of the LP/Farkas stack is required to assess reviewer-v4.

9. Provenance rule

Reviewer-v3 and all earlier packages remain frozen. The old computational route, the normalization bug, its correction and every audit report stay public. Reviewer-v4 is a successor, not a rewrite of history.

Hostile-audit disposition

The dedicated reviewer-v4 hostile audit found **no blocking flaw**. Its separately specified exact checker passed GitHub Actions run 34621982241. The audit report is [project/reviews/n29/2026-09-11-review](#). This is internal hostile review, not external validation.